

CS 331 (Software Engineering Lab)

Assignment 3

CI/CD Pipeline Automated Failure Recovery System

System Overview

The CI/CD Pipeline Automated Failure Recovery System is an intelligent software solution designed to automatically detect, diagnose, and recover from failures in continuous integration and continuous deployment pipelines. The system monitors pipeline executions, identifies failure patterns, applies recovery strategies, and learns from historical data to improve future recovery success rates.

Data Flow Diagrams (DFD)

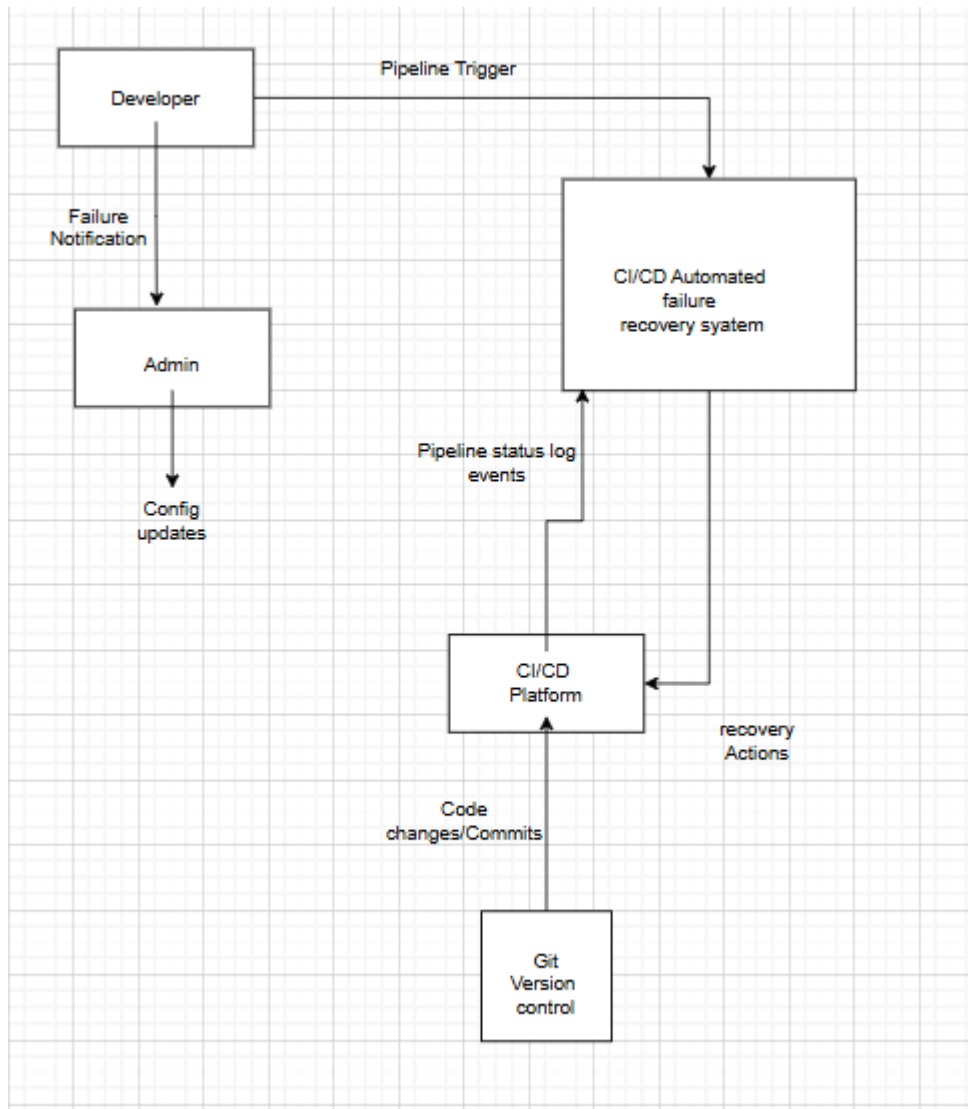
Q1. Context Diagram (Level 0 DFD)

The Context Diagram shows the CI/CD Failure Recovery System as a single process with external entities that interact with it.

External Entities:

- Developer: Triggers pipelines, receives notifications
- CI/CD Platform (Jenkins/GitLab): Executes pipelines, sends events
- Version Control System (Git): Provides code changes
- Admin: Configures recovery policies

Context Diagram Structure:



Data Flows:

- Pipeline Trigger: Developer → System (contains build parameters, branch info)
- Pipeline Status/Logs: CI/CD Platform → System (execution status, error logs)
- Code Changes: Version Control → System (commit history, file changes)
- Recovery Actions: System → CI/CD Platform (retry commands, configuration fixes)
- Failure Notifications: System → Developer (alert messages, recovery status)
- Config Updates: Admin → System (recovery policies, threshold settings)
- Recovery Reports: System → Admin (analytics, success rates)

Level 1 DFD

The Level 1 DFD breaks down the main system into detailed processes, data stores, and shows the data flow between them.

Processes:

- P1: Monitor Pipeline - Continuously monitors CI/CD pipeline execution
- P2: Detect Failure - Identifies and classifies pipeline failures
- P3: Analyze Root Cause - Diagnoses the underlying cause of failure
- P4: Select Recovery Strategy - Chooses appropriate recovery action
- P5: Execute Recovery - Implements the selected recovery strategy
- P6: Update Knowledge Base - Learns from recovery outcomes
- P7: Generate Reports - Creates analytics and notifications

Data Stores:

- D1: Pipeline Logs - Stores execution logs and events
- D2: Failure History - Records all past failures and patterns
- D3: Recovery Strategies - Contains recovery rules and actions
- D4: Configuration DB - Stores system settings and policies
- D5: Knowledge Base - Machine learning models and patterns

Level 1 DFD Structure:

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